DISASTER MANAGEMENT

METEROLOGICAL DISASTERS

MODULE - 7

DISASTER MANAGEMENT

DISASTERS ARISING FROM NATURAL HAZARDS

GEOPHYSICAL

- Earthquakes
- Volcano
- Tsunami

HYDROLOGICAL

- Avalanche
- Coastal Erosion
- Coastal Flood
- Hydrological flood
- Flash Flood
- Debris/mud flow

METEOROLOGICAL

- Cyclone
- Storm Surge
- Heat wave/ cold wave
- Heavy rains, Sand or dust storm

CLIMATOLOGICAL

- Droughts
- Extreme hot or cold conditions
- Forest fires
- Glacial lake outburst

BIOLOGICAL

- Epidemics – viral, bacterial, parasitic, fungal or prion infections
- Insect infestations
- Animal stampedes

DISASTER MANAGEMENT

CYCLONE RISK

1 In India, tropical cyclones occur in the months of May-June and October-November.

2 On an average, five to six tropical cyclones form every year, of which two or three could be severe.

3 The storm surges and torrential rainfall are the greatest killers of a cyclone, by which sea water inundates low lying areas of coastal regions and causes heavy floods, erodes beaches and embankments, destroys vegetation and reduces soil fertility.

4 As per broad scale assessment of the population at risk, nearly one third of India's population, is vulnerable to cyclone-related hazards. Climate change with the resultant sea-level rise and expected increase in severity of cyclones

India's long coastline of nearly 7,500 km

- 5,400 km along the mainland
- 132 km in Lakshadweep
- 1,900 km in the Andaman and Nicobar Islands

- About 10% of the World's tropical cyclones affect the Indian coast.
- More cyclones occur in the Bay of Bengal than in the Arabian Sea and the ratio is approximately 4:1.
- The cyclones of severe intensity and frequency in the northern part of the Indian Ocean are bi-modal in character, with their primary peak in November and secondary peak in May

CYCLONE RISK

BEST PRACTICES FOR VULNERABILITY MAPPING

1. Indian Meteorological Department(IMD) is the nodal agency for early warning of cyclones and floods.
2. Detailed maps to delineate coastal wetlands, mangroves and shelterbelts and tracts for coastal bio-shields
3. Studies on Ecosystem and shoreline changes and socio-economic impacts of cyclone and wind hazards
4. Carrying capacity monitoring social, physical, economic, ecological, gender, social inclusion and equity aspects
5. National Cyclone Risk Mitigation Project (NCRMP) launched by Home ministry to upgrade the forecasting, tracking and warning about cyclones in states.
6. Dissemination of warnings to all (including fishermen), down to the last mile
7. Providing warnings on radio, TV, and cell phones with seamless interface between national and state networks

CYCLONE RISK REDUCTION

UNDERSTANDING DISASTER RISK

- Short Term - Enhancement of Observational Network Stations (ONS), Automatic Weather Stations (AWS) and Rain-Gauge Network (RGN)
- Medium Term - Modernization of observation network, equipment, systems, technology , surge recorder for each coastal district, vulnerable to cyclones
- Long Term - Land- and Ocean-based observation systems , Research and studies to improve forecasts
- Observation Networks, Information Systems, Research, Forecasting, Early Warning responsibility of MOES and DOS, MOST, MEITY

INTERAGENCY COORDINATION

- Overall disaster governance MOES
- Organising and coordinating central assistance by MHA
- Warnings, Information, Data by MOES, MEITY, NDMA
- Revised /updated rules and norms for Non-structural measures MHA, NDMA, BIS
- SDMA, DDMA, Panchayats and ULBs will be responsible for coordinating the dissemination of warnings to all, down to the last mile – remote, rural or urban; Regular updates to people in areas at risk .
- State agencies are responsible for organising and coordinating.

INVESTING IN DRR – STRUCTURAL MEASURES

- Building Multi-Purpose Cyclone Shelters where NDMA, MHUA will provide Technical support whereas the state agencies will be involved in :
- Short term - Identification of safe buildings and sites to serve as temporary shelters for people and livestock evacuated from localities at risk
- Middle term - Construction of multi-purpose shelters in coastal villages/habitations prone to frequent cyclones .
- Hazard resistant construction, strengthening, and retrofitting of all lifeline structures.

CYCLONE RISK REDUCTION

INVESTING IN DRR – NON STRUCTURAL MEASURES

- The central agencies - MOES, MOEFCC, DOS, BIS, MFIN are involved in Guidance and Support , monitoring of compliance of coastal zone laws.
- The state agencies should be responsible for Ecologically sound land-use zonation; regulating aquaculture, and groundwater extraction; strengthen land-use planning.
- MOEFCC is involved in coastal shelterbelts as a mandatory component under national afforestation programme.
- Disaster Risk Audit for the future developmental project for both public and private entities.
- Guiding Public Private Partnerships where it is more people-centric.

CAPACITY DEVELOPMENT

- NIDM, NDMA, NLRTI is involved in training and orientation programs for central govt. staff, and other direct stakeholders.
- DMD, SDMA, DDMA, RD, SIDM is involved in Training and orientation programs for state govt. staff, civil society, media persons, veterinary care professionals.
- Training of NCC, NYKS, Scouts and Guides and NSS
- Update curriculum for undergraduate engineering courses to include topics relevant for cyclone Risk Management
- NDRF - Promoting the planning and execution of emergency drills
- Skill development for multi-hazard resistant construction

Learning Space

CHANGING CHARACTER OF CYCLONES

Odisha lost 10,000 people in the 1999 super cyclone. But during Cyclone fani in 2019, even when wind speeds reached 204 kilometre per hour, the loss of human life was contained to 41.

1. The emerging climate models predict increasing intensity and frequency of tropical cyclones.
2. As oceans warm, there is more moisture and ocean storms will build up to lash the land with devastating wind and rain.
3. IMD had nearly perfected the science of cyclone forecast, But during the recent cyclones of Ockhi, Titli and Fani, the technology is proved to be not accurate of clearly charting the paths of the cyclones
4. The recent cyclones are termed as rare or rarest of rare because they are not easily predictable by the scientific weather technologies.
5. The recent cyclones teach us that we must invest in the science of weather and in our governance capacity and it is better if we understand that the cyclones would become more unpredicable

DISASTER MANAGEMENT

HEAT WAVES

HEAT WAVE

1. Heat waves are a period of abnormally high temperatures that leads to physiological stress, which sometimes can claim human life.

2. The World Meteorological Organization defines a heat wave as five or more consecutive days during which the daily maximum temperature exceeds the average maximum temperature by five degrees Celsius.

3. Heat waves that are more intense in nature with each passing year, and have a devastating impact on human health thereby increasing the number of heat wave casualties.

DEFINED TERMINOLOGIES OF HEAT WAVES

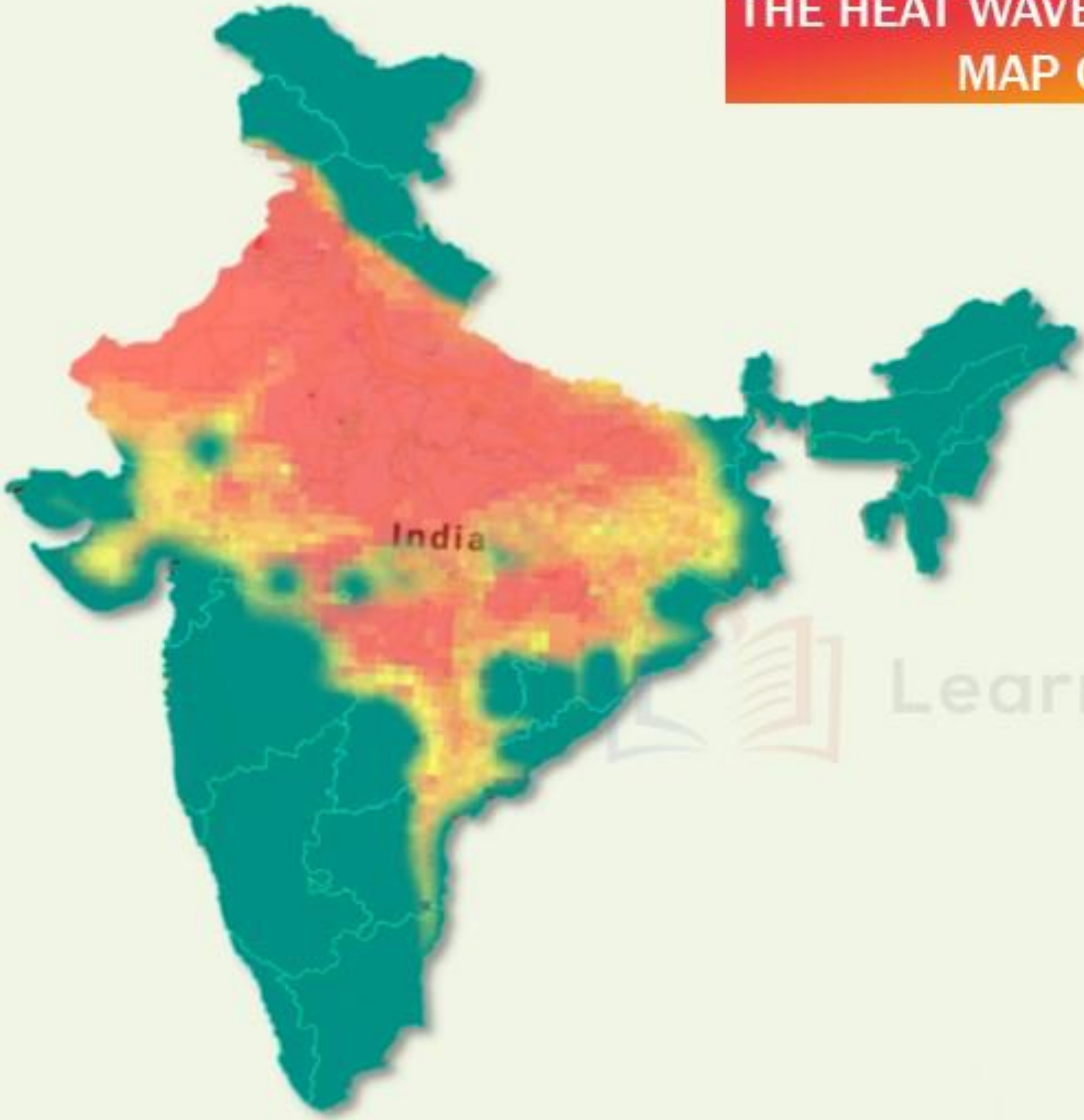
Heat wave is considered a "silent disaster" as it develops slowly

1 According to revised terminology of the IMD applicable from January 2016, in India, it will be considered as heat wave

- If the maximum temperature of a met-sub-station reaches at least 40°C or more in the plains
- 37°C or more in coastal areas
- At least 30°C or more for hilly regions.

2 IMD defines heat wave based on departure from Normal is 4.5°C to 6.4°C and Severe Heat Wave departure from normal is >6.4°C.

THE HEAT WAVES VULNERABILITY MAP OF INDIA



Heat Waves typically occur between March and June, and in some rare cases even extend until July.
Heat waves are more frequent over the Indo-Gangetic plains of India.

In the northern plains of the country, dust in suspension occurs in many years for several days, bringing minimum temperature much higher than normal and keeping the maximum temperature around or slightly above normal

EFFECTS OF HEAT WAVES

Heat Cramps: Edema (swelling) and Syncope (Fainting) generally accompanied by fever below $39^{\circ}C$ ($102^{\circ}F$)

Heat Exhaustion: Fatigue, weakness, dizziness, headache, nausea, vomiting, muscle cramps and sweating

Heat Stoke: Body temperatures of $40^{\circ}C$ (104oF) or more along with delirium, seizures or coma, which is a potentially fatal

WHY THE IPCC IS WARNING ABOUT UNPRECEDENTED HEAT WAVES

If all the countries are not serious about their emissions, the world can experience a 4°C rise in temperature by 2100

INTERGOVERNMENTAL PANEL ON climate change

WMO UNEP

1. In India, the warming is mainly contributed by the winter and post-monsoon seasons, which have increased by 0.80°C and 0.82°C in the last hundred years respectively.

2. A world with 4°C rise in temperature would be one of unprecedented heat waves, severe drought, and major floods in many regions, with serious impacts on ecosystems and associated services

3. India has been placed amongst the countries who most experience high social and economic costs from climate change

4. Magnified effect of heat waves can be seen due to paved and concrete surfaces in urban areas, a lack of tree cover, Urban heat island effects, Night time temperature raise.

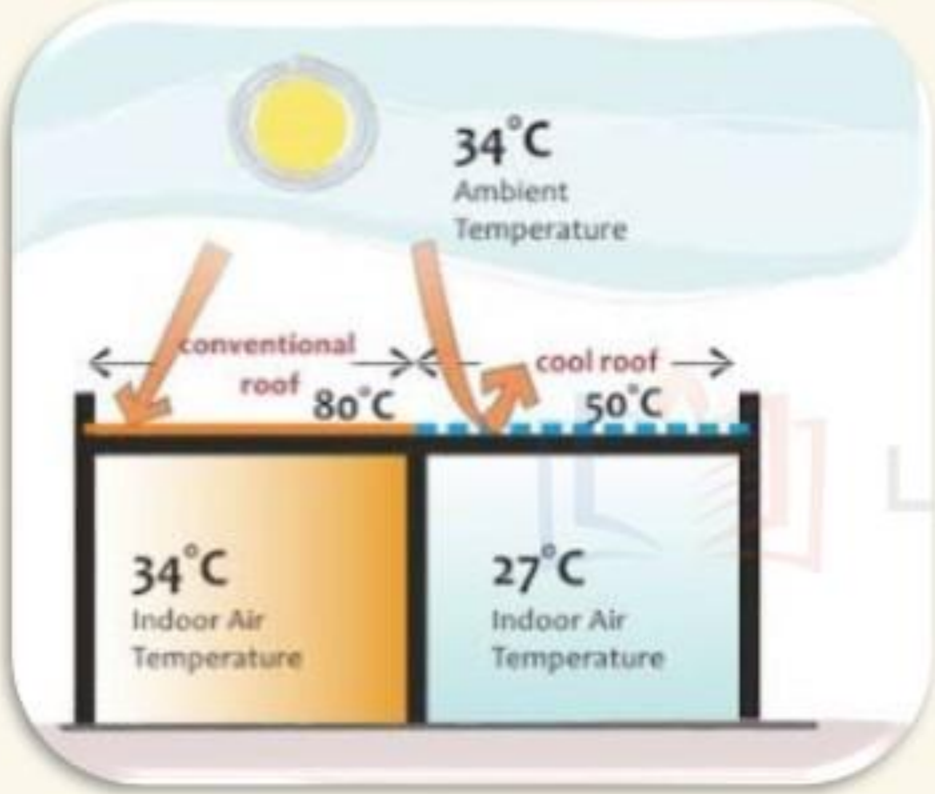
WHAT ARE THE WAYS IN WHICH HEAT WAVES SHOULD BE DEALT?

YELLOW ALERT	Hot Day Advisory	41.1˚C – 43˚C
ORANGE ALERT	Heat Alert Day	43.1˚C – 44.9˚C
RED ALERT	Extreme Heat Alert Day	≥ 45˚C

1. Identifying heat hot-spots through appropriate tracking of meteorological data
2. Implementation of local Heat Action Plans with strategic inter-agency co-ordination
3. Review of existing occupational health standards, labour laws and sectoral regulations for worker safety in relation to climatic conditions
4. Policy intervention and coordination across three sectors health, water and power , National Action Plan on Climate Change and Health

Learning Space

WHAT ARE THE WAYS IN WHICH HEAT WAVES SHOULD BE DEALT?



5. Research using sub-district level data to provide separate indices for urban and rural areas
6. Ensure an adequate supply of water. Timely access to drinking water can help mitigate this escalation
7. Popularisation of simple design features such as shaded windows, underground water storage tanks, passive cooling techniques, insulating housing materials
8. Advance implementation of local Heat Action Plans, plus effective inter-agency coordination to protect vulnerable groups and communities.

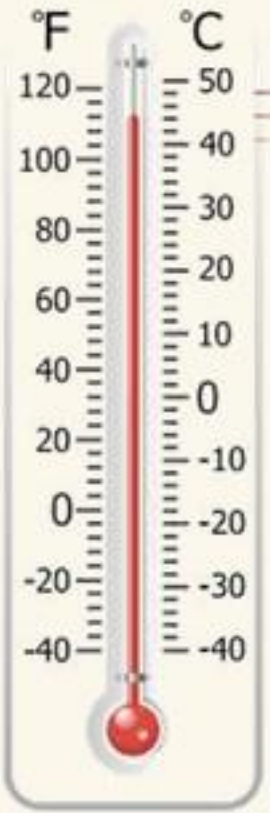
KEY AREAS OF HEAT ACTION PLANS


Dark rooftops in Delhi White rooftops in Jaisalmer

1. Building public awareness and community outreach to communicate the risks of heat waves
2. Initiating an early warning system and inter-agency coordination to alert residents of predicted high and extreme temperatures.
3. Special programs focused on heat-vulnerable communities, including programs for targeted communication and resilience efforts, such as providing drinking water and cool roofs.
4. Capacity building among health care professionals to recognize and respond to heat-related illnesses, particularly during extreme heat events.
5. Mapping high-risk areas, increasing outreach and communication in rural and semi arid areas, expanding access to potable drinking water stations and cooling spaces during extreme heat days.

CASE STUDY OF AHMEDABAD: BUILDING RESILIENCE SAVES LIVES

DAILY MAXIMUM TEMPERATURE



AHMEDABAD, A MAJOR INDIAN CITY, HAS AVOIDED MORE THAN **1,100 DEATHS ANNUALLY** SINCE THE CITY IMPLEMENTED SOUTH ASIA'S FIRST-EVER **HEAT ACTION PLAN** IN 2013.

TEMPERATURE THRESHOLD

- ≥45°C: 600+ deaths avoided
- 43–44.9°C: 300+ deaths avoided
- 41–42.9°C: 200+ deaths avoided

Since 2013, 23 states and over 100 cities and districts across India are developing and implementing life-saving heat action plans.

Authors: Jeremy J Hess, Sathish LM, Kim Knowlton, Shubhayu Saha, Priya Dutta, Parthasarathi Ganguly, Abhiyant Tiwari, Anjali Jaiswal, Perry Sheffield, Jayanta Sarkar, SC Bhan, Amit Begda, Tejas Shah, Bhavin Solanki, and Dileep Mavalankar

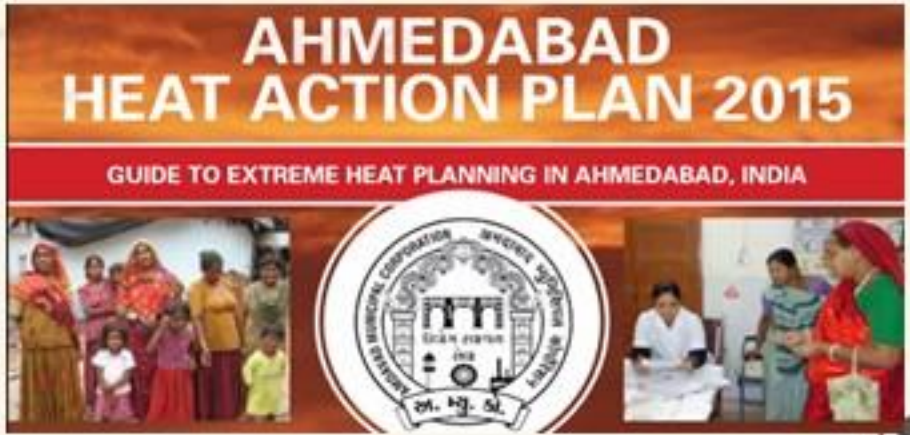
Hess JJ, et al. 2018. Building resilience to climate change: pilot evaluation of the impact of India's first Heat Activation Plan on all-cause mortality. Journal of Environmental and Public Health, vol. 2018, Article ID 7973519 (8 pp.). https://doi.org/10.1155/2018/7973519.

EARLY WARNING SYSTEM & INTER AGENCY EMERGENCY RESPONSE PLAN
Alert residents of predicted high and extreme temperatures & formally communication channels to alert governmental agencies

PUBLIC AWARENESS & COMMUNITY OUTREACH
Communicate the risks of heat waves and implement practices to prevent heat-related deaths and illnesses

CAPACITY BUILDING OF MEDICAL PROFESSIONALS
Training focus on primary medical officers and other paramedical staff, and community health staff

REDUCING HEAT EXPOSURE AND PROMOTING ADAPTIVE MEASURES
Access to potable drinking water and cooling spaces during extreme heat days & promote adaptive measures.



Learning Space

COLD WAVES IN INDIA

- The 2010 cloudburst in Leh, was caused due to the western disturbances.
- 2014, the Kashmir flood across many of its districts was also caused by the Western Disturbances.
- Many experts believe that the 2013 Uttarakhand floods may have occurred due to interactions between western disturbances and the summer monsoon

1. 'Core cold wave' zone covers Punjab, Himachal Pradesh, Uttarakhand, Delhi, Haryana, Rajasthan, Uttar Pradesh, Gujarat, Madhya Pradesh, Chhattisgarh, Bihar, Jharkhand, West Bengal, Odisha and Telangana.
2. Recently, the Indian Meteorological Department (IMD) has predicted India's 'core cold wave' zone is expected to experience higher minimum temperature during winters

India on an average is 0.5^{0} C warmer than 50 years ago

ANALYSIS OF WHY REPEATED COLD WAVES ARE GETTING FORMED?

1. Cold wave conditions form mainly due to western disturbances which are extratropical storms that mostly appear during the winter months - November to March

2. They originate somewhere near the Mediterranean sea and the Atlantic Ocean and are laden with moisture.

3. Western disturbances create precipitation and bring down day temperatures, but night temperatures remain steady.

4. Presence of an active western disturbance cold winds from the Himalayas continue to blow in to the northern parts of India thereby allowing the prevailing cold conditions to continue and intensify.

5. Once the western disturbances pass, day temperatures increase but night temperatures dip significantly. Parts of the plains of north-west, central and west India encounter cold wave, fog and ground frost conditions etc.

6. Exposure to extreme and especially unexpected cold can lead to hypothermia and frostbite, which can cause death and injury.

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WHAT ARE THE WAYS IN WHICH COLD WAVE AND FROST RISK REDUCTION CAN BE ACHIEVED?

1. As Cold Wave/Frost is a localized phenomenon, the relevant State Governments must draw up location specific mitigation plans involving respective DDMAs and local authorities (Panchayats and ULBs).

2. The State Governments must maintain close coordination with India Meteorological Department (MOES (IMD)) and closely monitor cold wave situation.

3. Warnings should be disseminated to the public through appropriate forums (including local newspapers and radio stations) on a regular basis

4. The MAFW(Ministry of Agriculture & Farmers' Welfare) closely monitors cold wave situation in consultation with MOES (IMD) and State Governments.

5. In case of cold wave/frost situation, States needs to initiate location specific measures as outlined in District Crop Contingency Plans and in consultation with respective State Agricultural Universities to minimize its impact.

Learning Space

RELATED QUESTIONS THAT CAN BE ANSWERED AFTER READING THE MODULE

1. What is heat wave? Why is India experiencing more heat waves? Also examine what strategy is being followed by states to address the consequences of heat waves.

2. India along with other tropical countries are experiencing more heat waves than ever before. What are the ways of decreasing the vulnerabilities of heatwaves.

3. Analyze the reasons behind the extra cold winters in parts of north India.

4. The Frequency of cyclones have decreased ever so slightly over the Indian Ocean Region but the intensity of such storms are rapidly increasing. Discuss the role of human-induced climate change in such events. What roles should the Central, State and Local authorities play to address such changes, increase resilience and Build Back Better?

WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

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